Payman Sharafianardakani

Robotics Research Engineer | Ph.D. Candidate | Full-Stack Roboticist

Louisville, KY • **Work Authorization: EAD Holder (Green Card Pending) – No Sponsorship Required**

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Professional Summary

Robotics engineer with 8 years of research and system-integration experience in physical human–robot interaction, safety-critical shared control, sensor fusion, and deep learning. Lead engineer on an NSF-funded clinical mobile manipulator, responsible for the whole stack: motor-controller PCBs and SLA-printed sensor hardware, ROS control and perception nodes, MPC-CBF safety filters, a browser teleoperation front end, and the IRB human-subject trials that validated all of it. Five peer-reviewed journal papers, and a hospital trial with patients and nurses.

Technical Skills

Languages: Python, C++, MATLAB, TypeScript/JavaScript, Bash

Robotics & Autonomy: ROS1 (Kinetic/Melodic/Noetic), ROS2 (Humble), Gazebo, RViz, Motion & Path Planning, MoveIt, ROS Navigation Stack (AMCL, costmaps), TF2, Kinova Kortex, Mecanum omnidirectional kinematics, URDF, NVIDIA Isaac Sim

Control & Optimization: PID/PD Loop Tuning (classical control), MPC, Control Barrier Functions (CBF-QP, MPC-CBF), OSQP, CasADi, Admittance/Impedance Control, Neuroadaptive Control, Lyapunov Stability Analysis, Genetic Algorithms (PyGAD), Real-Time Control (50–333 Hz loops), Kalman Filtering (Singer motion model), network-delay estimation

Perception & ML: PyTorch, OpenCV, TensorFlow, Keras, Scikit-learn, ONNX, CUDA, Semantic Segmentation (zero-shot, FastSAM), Temporal Convolutional Networks, BiLSTM, CNN-GRU, MLP, PointNet++, FastSAM, Contact-GraspNet, Reinforcement Learning, multi-modal tactile + force/torque fusion, LiDAR obstacle sectoring

Embedded & Hardware: Linux, real-time embedded firmware (RTOS), EtherCAT motor-control chains, CAN, SPI, I2C, UART, Arduino, Teensy, STM32 Nucleo, Sabertooth motor drivers, Jetson, VersaLogic single-board computers, analog signal conditioning, SolidWorks (DFM/assembly), FDM and SLA 3D printing, PCB design/debug/soldering, FSR tactile arrays (FlexiForce A201), ATI Axia 80 F/T sensors, multi-channel data acquisition (DAQ), RealSense RGB-D, 3D LiDAR

Software & Infrastructure: distributed system architecture, CI/CD (GitHub Actions), Next.js (App Router), React, TypeScript, Tailwind CSS, static-export SPA, WebSockets, rosbridge, Cloudflare Tunnel + Access (SSO), CBOR/JPEG stream compression, socket programming, Docker, Git

Professional Experience

Robotics Research Engineer (Ph.D. Research Assistant) — Sep. 2022 – Present
Louisville Automation & Robotics Research Institute (LARRI), University of Louisville — *Louisville, KY*

- Lead engineer on ARNA, the Adaptive Robotic Nursing Assistant funded by the NSF Future of Work program (#FW-HTF-RM): an omnidirectional mobile manipulator (a 7-DOF Kinova Gen3 arm on a Mecanum base) deployed on an active hospital unit. Responsible for its control, perception, embedded firmware, web infrastructure, and clinical trials.
- Built ARNA's complete remote-operation stack: network-aware safety filtering, secure browser transport, and one-click autonomous grasping. Validated it with **30 remote operators** across multiple U.S. cities, with every command crossing the public internet.
  - Designed a five-layer shared-control stack for remote teleoperation that holds hard safety guarantees when *both* the operator and the connection are unreliable. Adaptive layers can only make the robot more cautious; barrier constraints are never relaxed. A 100 Hz arm MPC-CBF gave **zero boundary violations in 60 runs** at jerk cost **200–440**$\times$ **lower** than raw operator input; a 50 Hz OSQP base filter on LiDAR sectors gave zero margin violations; a watchdog scaled margins as the link degraded, with a 3.0 s dwell before recovery.
  - Wrote a TypeScript/Next.js teleoperation interface an operator opens as a URL. Operators install nothing, and the robot exposes no inbound port; access runs over an outbound Cloudflare Tunnel with identity-based Access. Compiled it to a fully static bundle, so no Node runtime runs on the robot. Split the session across three isolated rosbridge instances so camera bursts cannot head-of-line-block a joystick or stop command, and cut per-frame data volume $\sim$**35–40**$\times$ with source-side JPEG encoding and CBOR subscriptions.
  - Cut remote grasping to one click on the target object: FastSAM segments the clicked object on CPU, and Contact-GraspNet with a PointNet++ encoder proposes ranked SE(3) grasps. Deployed it on a **4 GB laptop GPU** by cropping inference to the object bounding region ($\sim$10$\times$ input reduction) and warm-loading weights once at node start. Ranked grasp candidates by reachability instead of network confidence, which is what actually limited success.
- Led the sensorized handlebar and real-time adverse-event detector end to end: mechanical design, electronics, ML, and human trials. Built it around **eight FlexiForce A201 FSRs** and a custom Arduino signal-conditioning board publishing at **122 Hz**, fused with 6-axis F/T through a Singer-model Kalman filter. Benchmarked TCN vs. BiLSTM vs. CNN-GRU under subject-wise cross-validation, then deployed the TCN to ONNX onboard the robot, with no cloud, re-evaluating every **8.2 ms**. On **10 unseen participants**: **85.0% detection** at **2.05 s mean latency**.
- Learned the joystick-to-robot teleoperation mapping as a black box, with no kinematic model of either device: two networks in parallel, one searched offline by a genetic algorithm to minimize task time, one trained online by SGD against measured jerk. In a 20-subject IRB study, **linear jerk halved** and **angular jerk fell** $\sim$**3**$\times$ ($p < .001$), **with no increase in task-completion time**.

- Designed a model-free neuroadaptive admittance controller that learns ARNA's unmodelled dynamics online, with update laws derived from a Lyapunov analysis, running at 333 Hz against a 125 Hz admittance model. Against a tuned PD baseline with 10 users: tracking accuracy improved up to **32%** and motion jerk fell up to **52%**. Re-tuned with **63 nursing students**: velocity tracking error **0.0157** $\rightarrow$ **0.0105 m/s**, mean actuator torque **3.66** $\rightarrow$ **1.79 Nm**.
- Designed and led an acceptability trial on an active unit at University of Louisville Hospital with **10 patients and 5 nurses**. Paired objective task-time logging with Technology Acceptance Model instruments, analyzed with linear mixed-effects models in R: the joystick was significantly easier to use and more likely to be adopted than the tablet ($p \approx 0.02$–0.03) *despite statistically similar task times*.
- Repaired and upgraded ARNA's motor-controller PCBs, brought non-functional sonar and bump sensors back online and into ROS, repurposed the LiDAR for navigation, and rebuilt onboard wiring across the EtherCAT motor chain, Sabertooth motor drivers, Teensy microcontrollers, battery/inverter bank, and instrumentation board. Worked across the robot's serial and fieldbus layers (EtherCAT, CAN, SPI, I2C, and UART) to bring sensors and drives back under ROS control.
- Maintain and extend ARNA's embedded layer: a custom motor-controller PCB carrying **two Teensy microcontrollers** that must stay synchronised to drive the Sabertooth motor drivers coherently, speaking EtherCAT to the robot's main PC, plus an STM32 Nucleo handling the safety sensors. Maintained, debugged and repaired this firmware across several hardware revisions.
- Mentored **one M.S. student and six undergraduate researchers**, and authored the lab manuals and Standard Operating Procedures for ARNA that new researchers onboard from.

Mechatronics Engineer (M.Sc. Research Assistant) — Nov. 2017 – Sep. 2021
University of Tehran – Human-Robot Interaction Lab — *Tehran, Iran*

- Wrote a sound-source localization algorithm for an interactive social robot used in autism therapy. Audio filtering improved **70%** and gesture detection accuracy rose **40%**.
- Designed the "Baby Tank" open-source mobile platform, wrote its navigation and path-planning algorithms, and integrated the low-level microcontroller for active damping.
- Built a cross-platform Activity Reminder GUI in C++ and Qt Creator, to accessibility standards.

Education

Ph.D. Candidate, Electrical & Robotics Engineering, University of Louisville	Louisville, KY	Sep. 2022 – **Dec. 2026 (expected)**
M.S. in Mechatronics Engineering, University of Tehran	Tehran, Iran	Nov. 2017 – Sep. 2021
B.Sc. in Electrical Power Engineering, Islamic Azad University	Varamin, Iran	Sep. 2012 – Aug. 2017

Selected Publications

Full list: Google Scholar

- **P. Sharafianardakani**, M. A. Hanafy, I. Kondaurova, A. Ashary, M. M. Rayguru, D. O. Popa. "Adaptive User Interface With Parallel Neural Networks for Robot Teleoperation." *IEEE Robotics and Automation Letters*, 10(2):963–970, 2025.
- **P. Sharafianardakani**, S. Abubakar, S. K. Das, S. Surupa, M. A. Hanafy, M. Rayguru, D. O. Popa. "Evaluation of a Neuroadaptive Admittance Controller for Ambulation through Physical Human–Robot Interaction." *Intelligent Service Robotics*, 18(7):1355–1368, 2025.
- **P. SharafianArdakani**, I. Kondaurova, A. Ashary, N. Zhang, M. C. Logsdon, M. Walker, D. O. Popa. "Hospital Trial Results for the Acceptability of an Adaptive Robot Nursing Assistant." *IEEE CASE*, pp. 2043–2049, 2025.
- I. Kondaurova, **P. Sharafian**, R. Mitra, M. M. Rayguru, *et al.*, D. O. Popa. "Acceptance of an Adaptive Robotic Nursing Assistant for Ambulation Tasks." *Robotics* (MDPI), 14(9):121, 2025.
- M. C. Logsdon, I. Kondaurova, N. Zhang, *et al.* (incl. **P. SharafianArdakani**), D. O. Popa. "Perceived Usefulness of Robotic Technology for Patient Fall Prevention." *Workplace Health & Safety*, 72(12):542–549, 2024.
- **P. Sharafianardakani** *et al.* "Tactile Handlebar and Deep Learning for Safe Interaction with a Robot Nursing Assistant." *IEEE Transactions on Medical Robotics and Bionics* – **under review**.

Leadership & Awards

Journal Reviewer, IEEE Robotics and Automation Letters (RA-L), 2024–2025 • **Conference Reviewer**, ACM/SIGAPP SAC, 2024 • **STEM Mentor**, RAMPS Program – 30+ students, 2023–Present

Dissertation Completion Award (full tuition + stipend), University of Louisville, 2026 • **IEEE RAS Travel Support Award** (ICRA 2025) • **Graduate Student Travel Award**, 2025